WACC/NPV RECAP

- **A business is planning to invest K80m equity finance and K40m debt finance is a 4 year project. The interest on debt is 10% while the shareholders required rate of return is 15%. The expected before tax net cash inflows of the project will be;**

Year	**1**	**2**	**3**	**4**
Net cash inflows (K'm)	**70**	**120**	**90**	**50**

- **Company tax is 25% per year payable in the same year as profits. Capital allowances are available on a straight basis. The project will have a nil scrap value at 4 years.**
- **<u>Required:</u> Assess the viability of this project based on NPV**

WACC

Capital element	Cost	Value (K'm)	Proportion	Cost * Prop
Equity	15%	80	80/120	10.00
Debt	10%*(1-0.25)	40	40/120	2.50
			WACC	12.50%

- Capital Allowances = K120m/4 = K30m per year
- Tax shield due to capital allowances (K30m*0.25) = K7.5m per year (yr 1 to 4)

WACC/NPV (K'M)

Year	0	1	2	3	4
Cash inflows		50.0	80.0	70.0	40.0
Tax at 25%		(12.5)	(20.0)	(17.5)	(10.0)
Tax		7.5	7.5	7.5	7.5
Initial cost	(120.0)				
Net cash flows	(120.0)	45.0	67.5	60.0	37.5
Discount at 12.5%	1.000	0.889	0.790	0.702	0.624
Present Values	(120.0)	40.0	53.3	42.1	23.4
NPV	38.8				
Comment:	The project is viable since npv is positive.				

Adjusted Present Value (APV)

- **The APV method is an approach that evaluates the project and the impact of the financing separately.**
- **Used if a new project has a different financial risk (debt to equity ratio) from the company, i.e., the overall capital structure of the company changes.**

Adjusted Present Value (APV)

- ***APV consists of two different decisions:***
- **APV = Investment decision + Financing decision**
- **Value of a geared project = Value of an all-equity financed project + Present value of financing side effects.**
- **Value of an all equity financed project = the base case NPV.**

APV-Investment decision

- **The project is evaluated as though it were being undertaken by an all equity company with all financing side effects ignored.**
- ***Therefore:***
  - **Once the base case NPV is identified, the PV of the financing side effects is evaluated.**
  - **Use a beta that reflects just the business risk, i.e. asset beta.**
  - **Asset beta is used to get the cost of equity used in the base case NPV calculation.**

APV-Financing decision

- ***Financing cash flows consist of:***
  - **Issue costs (of equity and or debt)**
  - **Tax reliefs (savings).**
- ***As all financing cash flows are low risk (mostly predictable) they are discounted at either:***
  - **The cost of debt or**
  - **The risk free rate.**
- **PV of financing cash flows is then added to the base case NPV to find the APV.**

Grossing up

- **Where issue costs of finance exist, there is need to gross up the amount required so as to know the full amount needed to be raised *(inclusive of the issue costs).***
- **<u>Practice Question:</u>**
- ***The finance required for a planned investment is K2.4m (net of issue costs). Issue costs are 4%. And the finance raised will also have to cover the issue costs. What are the issue costs and what sum will need to be raised altogether?***

Grossing up and issue costs

- **The K2.4m is 96% of the amount to be raised:**
- **Therefore, (K2.4m/0.96) = K2,500,000 will be needed.**
- ***Issue costs are 4%***
- **4% × K2,500,000 = K100,000**
- ***Issue costs can be calculated in one stage as:***
- **K2.4m × 4/96 = K100,000**
- ***Issue costs of equity are NOT tax deductible.***
- ***Issue costs of debt are tax deductible.***

Calculating APV.

- ***The base case NPV is used as a starting point. The costs and benefits of the financing are then added to find a final adjusted present.***
- **Base case NPV** **X**
- **<u>PV of the issue costs:</u>**
  - **Equity** **(X)**
  - **Debt** **(X)**
- **PV of the tax shield** **<u>X</u>**
- **Adjusted present value** **<u>X</u>**

Practice question

- **Noble plc has a project, the cash flows are**
- **Year 0 1 2 3 4**
- **Cash flows K (10,000) 1,500 2,500 4,500 5,000.**
- **The ungeared cost of equity is 10%.**
- **<u>Required:</u> *Calculate the base case NPV and comment on the results***
- **If the firm use K6,000 of 8% debt to finance the project and face a tax rate of 30%. The debt will be reduced by K1,500 at the end of each year. Net of tax issue cost is K237.**

What is the APV of the project?

Base case NPV

Year	Cash flows K	DF at 10%	Present value K
0	(10,000)	1.000	(10,000)
1	1,500	0.909	1,364
2	2,500	0.826	2,065
3	4,500	0.751	3,380
4	5,000	0.683	3,415
		Base case NPV	224

Present value of interest tax shield (PVITS)

Year	Loan at Start K	Repmt K	Interest at 8% K	Tax shield at 30% K	Df at 8%	PVITS K
1	6,000	1,500	480	144	0.926	133
2	4,500	1,500	360	108	0.857	93
3	3,000	1,500	240	72	0.794	57
4	1,500	1,500	120	36	0.735	26
					Total	309

Calculating APV.

	K
Base case NPV	224
PV of the issue costs:	
Debt	(237)
PV of the tax shield	309
Adjusted present value	296

Decision: Project is acceptable because the APV is positive.